

Helix OTA “Issues (open workable items)

Revision: 1 **Last modified:** 2026-06-10T16:00:00Z

This is the canonical open-work tracker (Â§11.4.15 Status, Â§11.4.16 Type, Â§11.4.54 ATM-NNN). Closed items migrate to [Fixed.md](#). The short-form companion is [Issues_Summary.md](#). Items surfaced during the 2026-06-10 emulator-driven device-testing session; HEAD at creation a839220 + the docs commit on top.

Â§1. [ATM-001] Telemetry deployment_id not derivable from ota_protocol.UpdateAvailable

Status: In progress **Type:** Bug

A real device has no protocol-level way to obtain the deployment_id it must echo back in its telemetry. The server requires it (server/internal/api/handlers_client.go), but the protocol payload a device receives (submodules/ota-protocol/types.go, UpdateAvailable) carries no deployment_id field “ so the device cannot supply a value it was never told. Surfaced by the Tier-1 Go device-emulator (7dc3334) when it tried to round-trip register “ update-check “ telemetry.

A sibling agent is implementing the protocol-completeness fix this session (adding the field to the ota-protocol UpdateAvailable payload + wiring it through the server’s update-check response). This tracker entry stays In progress until that fix lands with the RED “ GREEN evidence under docs/qa/<run-id>/.

Â§2. [ATM-002] No GET /deployments list endpoint to enumerate deployments

Status: In progress **Type:** Feature

server.go exposes only POST /deployments (create) and GET /deployments/{id} (fetch one). There is no list endpoint to enumerate existing deployments, so neither the dashboard nor an operator/automation client can discover deployment IDs without already knowing them. A sibling agent is adding the GET /deployments list endpoint (with the same cursor-pagination convention as the group/members endpoints shipped in 50ef5c6) this session.

Â§3. [ATM-003] Emulator Tier-2 “ real Android A/B (update_engine/AVB/dm-verity auto-rollback) is host-gated

Status: Operator-blocked **Type:** Task

Operator-Block-Details: - **WHAT:** Stand up the Tier-2 emulator “ a real Android A/B update_engine payload-apply with AVB/dm-verity verification and auto-rollback, driven end-to-end against the control plane (per docs/design/

EMULATED_DEVICE_TESTING.md). - **WHY:** Tier-2 requires Cuttlefish (cvd) on a Linux host with nested KVM. Self-resolution exhausted (Â§11.4.21): (a) no host CLI/virtualisation path â€” the current host is Apple-Silicon with applehv, which cannot run Cuttlefish or nested KVM; (b) subagent delegation cannot conjure the missing kernel virtualisation; (c) the containers submodule (podman/applehv) cannot host a KVM-backed Cuttlefish guest on this hardware; (d) no captured fallback substitutes for real update_engine partition behaviour; (e) external research confirms Cuttlefishâ€™s Linux+KVM requirement â€” not a tooling gap. NOT structurally impossible (Â§11.4.112) â€” host/hardware-gated only. - **UNBLOCK CONDITION:** Access to a Linux host (or CI runner) with nested-KVM enabled where Cuttlefish (cvd) boots. - **WHO:** Operator â€” provision the Linux+KVM host / CI runner; see docs/design/EMULATED_DEVICE_TESTING.md Tier-2.

Â§4. [ATM-004] Emulator Tier-3 â€” real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real partitions

Status: Operator-blocked **Type:** Task

Operator-Block-Details: - **WHAT:** Tier-3 validation on the physical board â€” real vendor HAL, U-Boot A/B slot-switch, and dm-verity over real partitions, exactly as a fielded device runs (per docs/design/EMULATED_DEVICE_TESTING.md). - **WHY:** Requires the physical RK3588 / Orange Pi 5 Max hardware, which is not attached to this host. Self-resolution exhausted (Â§11.4.21): (a) no remote board over the available CLIs/ADB/SSH; (b) subagent delegation cannot substitute for absent silicon; (c) no repo tooling emulates real U-Boot slot-switch + on-silicon dm-verity; (d) no captured fallback reproduces real-partition AVB rollback; (e) research confirms the vendor HAL + bootloader path is board-specific. NOT structurally impossible (Â§11.4.112) â€” hardware-gated only. - **UNBLOCK CONDITION:** A physical RK3588 / Orange Pi 5 Max board reachable over ADB/SSH (or physically attached) for flashing. - **WHO:** Operator â€” attach / provide remote access to the board.